Quadratic Equations

QUADRATIC EQUATIONS

Solutions

Series K

EVERYTHING YOU NEED TO KNOW

100%

AND NOTHING MORE

MATHLETICS

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Page 3 questions

What is the difference between a linear and a quadratic equation?

In a linear equation the highest power of the variable (eg x) has a power of 1. In a quadratic equation the highest power of the variable has a power of 2 (eg x^2).

Find the solutions to the variable in each of these equations. Leave the solutions in surd form if necessary, and state if each equation has one, two or no real solutions:

a
$$x^2 = 4$$
$$x^2 = (\pm 2)^2$$
$$x = \pm 2$$

b
$$2b^2 = 32$$
$$b^2 = 16$$
$$b^2 = (\pm 4)^2$$
$$b = \pm 4$$

c
$$3b^2 - 75 = 0$$
$$b^2 = 25$$
$$b^2 = (\pm 5)^2$$
$$b = \pm 5$$

d
$$6y^2 = 0$$
$$y^2 = 0$$
$$y = 0$$

e
$$4t^2 - 28 = 0$$
$$t^2 = \frac{28}{4}$$
$$t^2 = (\pm\sqrt{7})^2$$
$$t = \pm\sqrt{7}$$

f
$$h^2 = -9$$
no real solutions

g
$$2p^2 = -8$$
$$p^2 = -4$$
no real solutions

h
$$8m^2 + 5 = 5$$
$$8m^2 = 0$$
$$m = 0$$

i
$$9x^2 - 16 = 0$$
$$x^2 = \frac{16}{9}$$
$$x^2 = \left(\pm\frac{4}{3}\right)^2$$
$$x = \pm\frac{4}{3}$$

j
$$-3k^2 + 108 = 0$$
$$k^2 = 36$$
$$k^2 = (\pm 6)^2$$
$$k = \pm 6$$

Page 5 questions

Rewrite these polynomials in the form $ax^2 + bx + c = 0$. Identify the values of a, b and c:

a $x^2 + 3x + 2 = 0$

$a = 1,\ b = 3,\ c = 2$

b $2x^2 + 4x + 5 = 0$

$a = 2,\ b = 4,\ c = 5$

c $x^2 - 7 = 0$

$x^2 + 0x - 7 = 0$

$a = 1,\ b = 0,\ c = -7$

d $x(x + 4) = 0$

$x^2 + 4x + 0 = 0$

$a = 1,\ b = 4,\ c = 0$

e $3x(4x - 5) = 0$

$12x^2 - 15x + 0 = 0$

$a = 12,\ b = -15,\ c = 0$

f $(x + 3)(x - 7) = 0$

$x^2 + 3x - 7x - 21 = 0$

$x^2 - 4x - 21 = 0$

$a = 1,\ b = -4,\ c = -21$

g $(3x + 5)(x - 8) = 0$

$3x^2 + 5x - 24x - 40 = 0$

$3x^2 - 19x - 40 = 0$

$a = 3,\ b = -19,\ c = -40$

h $-3(x + 4)(x - 1) = 0$

$-3(x^2 + 4x - x - 4) = 0$

$-3x^2 - 9x + 12 = 0$

$a = -3,\ b = -9,\ c = 12$

Solve these quadratic equations for the missing variable:

a $x(x - 2) = 0$

either $x = 0$ or $x - 2 = 0$ (by Null Factor Law)

$x = 0$ or $x = 2$

b $x(x + 4) = 0$

$x = 0$ or $x + 4 = 0$ (by Null Factor Law)

$x = 0$ or $x = -4$

c $x(2x + 1) = 0$

$x = 0$ or $2x + 1 = 0$ (by Null Factor Law)

$x = 0$ or $x = -\frac{1}{2}$

d $(x - 3)(x + 5) = 0$

$x - 3 = 0$ or $x + 5 = 0$ (by Null Factor Law)

$x = 3$ or $x = -5$

e $y(2y - 3) = 0$

$y = 0$ or $2y - 3 = 0$ (by Null Factor Law)

$y = 0$ or $y = \frac{3}{2}$

f $(2x + 7)(3x - 8) = 0$

$2x + 7 = 0$ or $3x - 8 = 0$

$x = -\frac{7}{2}$ or $x = \frac{8}{3}$

Page 6 questions

Solve these quadratic equations by factorising:

a $x^2 + 8x = 0$

$x(x+8) = 0$

$x = 0$ or $x + 8 = 0$ (by Null Factor Law)

$x = 0$ or $x = -8$

b $4x^2 + 12x = 0$

$4x(x+3) = 0$

$4x = 0$ or $x + 3 = 0$ (by Null Factor Law)

$x = 0$ or $x = -3$

c $t^2 - 6t = 0$

$t(t-6) = 0$

$t = 0$ or $t - 6 = 0$ (by Null Factor Law)

$t = 0$ or $t = 6$

d $5b^2 - 15x = 0$

$5b(b-3) = 0$

$5b = 0$ or $b - 3 = 0$ (by Null Factor Law)

$b = 0$ or $b = 3$

e $y^2 - 4y - 21 = 0$

$(y-7)(y+3) = 0$

$y - 7 = 0$ or $y + 3 = 0$
(by Null Factor Law)

$y = 7$ or $y = -3$

f $m^2 - m - 30 = 0$

$(m-6)(m+5) = 0$

$m - 6 = 0$ or $m + 5 = 0$
(by Null Factor Law)

$m = 6$ or $m = -5$

Page 7 questions

Solve these quadratic equations by factorising:

g $2n^2 + 8n - 64 = 0$

$2n^2 + 8n - 64 = 0$

$2(n^2 + 4n - 32) = 0$

$2(n + 8)(n - 4) = 0$

$n + 8 = 0$ or $n - 4 = 0$
(by Null Factor Law)

$n = -8$ or $n = 4$

h $x^2 + 5x = 6$

$x^2 + 5x - 6 = 0$

$(x - 1)(x + 6) = 0$

either $x - 1 = 0$ or $x + 6 = 0$
(by Null Factor Law)

$x = 1$ or $x = -6$

i $3x(x + 2) = 105$

$3x^2 + 6x - 105 = 0$

$3(x^2 + 2x - 35) = 0$

$3(x + 7)(x - 5) = 0$

either $x + 7 = 0$ or $x - 5 = 0$
(by Null Factor Law)

$x = -7$ or $x = 5$

j $(p + 3)^2 = 8p + 72$

$p^2 + 6p + 9 = 8p + 72$

$p^2 - 2p - 63 = 0$

$(p - 9)(p + 7) = 0$

$p - 9 = 0$ or $p + 7 = 0$
(by Null Factor Law)

$p = 9$ or $p = -7$

Page 9 questions

Complete the square of the following trinomials:

1. $x^2 + 4x + 1 = (x+2)^2 - 2^2 + 1$

$= (x+2)^2 - 3$

2. $x^2 - 6x + 16 = (x-3)^2 - (-3)^2 + 16$

$= (x-3)^2 + 7$

3. $x^2 + 3x + 7 = \left(x + \frac{3}{2}\right)^2 - \left(\frac{3}{2}\right)^2 + 7$

Hint : $\left(\frac{3}{2}\right)^2 = \frac{9}{4}$

$= \left(x + \frac{3}{2}\right)^2 - \frac{9}{4} + 7$

$= \left(x + \frac{3}{2}\right)^2 - \frac{9}{4} + \frac{28}{4}$

$= \left(x + \frac{3}{2}\right)^2 + \frac{19}{4}$

4. $3x^2 + 24x + 30 = 3[x^2 + 8x + 10]$

$= 3[(x+4)^2 - 4^2 + 10]$

$= 3[(x+4)^2 - 6]$

$= 3(x+4)^2 - 18$

5. $-x^2 + 10x - 23 = -[x^2 - 10x + 23]$

Hint: factorise -1 out using square brackets

$= -[(x-5)^2 - (-5)^2 + 23]$

$= -[(x-5)^2 - 2]$

$= -(x-5)^2 + 2$

6. $-3x^2 - 6x + 18 = -3[x^2 + 2x - 6]$

$= -3[(x+1)^2 - 1^2 - 6]$

$= -3[(x+1)^2 - 7]$

$= -3(x+1)^2 + 21$

Page 11 questions

Solve for x in the following:

a $(x+2)^2 = 5$

$$(x-2)^2 = \pm(\sqrt{5})^2$$
$$x+2 = \pm\sqrt{5}$$
so $x+2 = \sqrt{5}$ or $x+2 = -\sqrt{5}$
$$x = -2+\sqrt{5} \text{ or } x = -2-\sqrt{5}$$

b $(x-3)^2 = 16$

$$(x-3)^2 = (\pm 4)^2$$
$$x-3 = \pm 4$$
$$x = 3-4 \text{ or } x = 3+4$$
$$x = -1 \text{ or } x = 7$$

c $3(x-7)^2 = 8$

$$(x-7)^2 = \frac{8}{3}$$
$$x-7 = \left(\pm\sqrt{\frac{8}{3}}\right)^2$$
$$x-7 = \left(\pm\sqrt{\frac{4\times 2}{3}}\right)^2$$
$$x = 7+2\sqrt{\frac{2}{3}} \text{ or } x = 7-2\sqrt{\frac{2}{3}}$$

d $-5(x+6)^2 = 35$

$$(x+6)^2 = -7$$
no real solution

Complete the square and then solve for x in the following:

a $x^2 + 6x + 2 = 0$

$$(x+3)^2 - 3^2 + 2 = 0$$
$$(x+3)^2 = 7$$
$$(x+3)^2 = (\pm\sqrt{7})^2$$
$$x+3 = \pm\sqrt{7}$$
$$x = -3+\sqrt{7} \text{ or } x = -3-\sqrt{7}$$

b $x^2 - 10x + 15 = 0$

$$(x-5)^2 - (-5)^2 + 15 = 0$$
$$(x-5)^2 = 10$$
$$(x-5)^2 = (\pm\sqrt{10})^2$$
$$x-5 = \pm\sqrt{10}$$
$$x = 5+\sqrt{10} \text{ or } x = 5-\sqrt{10}$$

Page 12 questions

Complete the square and solve for the variable in the following:

a $2q^2 + 3q - 2 = 0$

$$2\left[q^2 + \frac{3}{2}q - 1\right] = 0$$

$$2\left[\left(q + \frac{3}{4}\right)^2 - \left(\frac{3}{4}\right)^2 - 1\right] = 0$$

$$2\left[\left(q + \frac{3}{4}\right)^2 - \left(\frac{9}{16}\right) - \left(\frac{16}{16}\right)\right] = 0$$

$$\left(q + \frac{3}{4}\right)^2 - \frac{25}{16} = 0 \text{ (divide both sides by 2)}$$

$$\left(q + \frac{3}{4}\right)^2 = \left(\pm\frac{5}{4}\right)^2$$

$$q = -\frac{3}{4} + \frac{5}{4} \text{ or } q = -\frac{3}{4} - \frac{5}{4}$$

$$q = \frac{1}{2} \text{ or } q = -2$$

b $-m^2 - 2m + 5 = 0$

$$-[m^2 + 2m - 5] = 0$$

$$-[(m + 1)^2 - 1^2 - 5] = 0$$

$$(m + 1)^2 - 6 = 0$$

(divide both sides by negative -1)

$$(m + 1)^2 = (\pm\sqrt{6})^2$$

$$m + 1 = \pm\sqrt{6}$$

$$m = -1 + \sqrt{6} \text{ or } m = -1 - \sqrt{6}$$

c $4t^2 + 8t - 1 = 0$

$$4\left[t^2 + 2t - \frac{1}{4}\right] = 0$$

$$4\left[(t + 1)^2 - 1^2 - \frac{1}{4}\right] = 0$$

$$(t + 1)^2 - \frac{5}{4} = 0 \text{ (divide both sides by 4)}$$

$$(t + 1)^2 = \left(\pm\sqrt{\frac{5}{4}}\right)^2$$

$$t + 1 = \pm\sqrt{\frac{5}{4}}$$

$$t = -1 + \sqrt{\frac{5}{4}} \text{ or } t = -1 - \sqrt{\frac{5}{4}}$$

d $-3x^2 + 12x - 2 = 0$

$$-3\left[x^2 - 4x + \frac{2}{3}\right] = 0$$

$$-3\left[(x - 2)^2 - (-2)^2 + \frac{2}{3}\right] = 0$$

$$(x - 2)^2 - 4 + \frac{2}{3} = 0 \text{ (divide both sides by } -3\text{)}$$

$$(x - 2)^2 = \frac{12}{3} - \frac{2}{3}$$

$$(x - 2)^2 = \left(\pm\sqrt{\frac{10}{3}}\right)^2$$

$$x = 2 + \sqrt{\frac{10}{3}} \text{ or } x = 2 - \sqrt{\frac{10}{3}}$$

Page 13 questions

Consider the equation $ax^2 + bx + c = 0$ where a, b and c are any constants:

a Complete the square for this equation

$$ax^2 + bx + c = 0$$

$$a\left[x^2 + \frac{b}{a}x + \frac{c}{a}\right] = 0$$

$$a\left[\left(x + \frac{b}{2a}\right)^2 - \left(\frac{b}{2a}\right)^2 + \frac{c}{a}\right] = 0$$

$$\left(x + \frac{b}{2a}\right)^2 - \left(\frac{b}{2a}\right)^2 + \frac{c}{a} = 0 \quad \text{(divide both sides by } a\text{)}$$

$$\left(x + \frac{b}{2a}\right)^2 = \frac{b^2}{4a^2} - \frac{c}{a}$$

b Solve for x

$$\left(x + \frac{b}{2a}\right)^2 = \frac{b^2}{4a^2} - \frac{c}{a}$$

$$x + \frac{b}{2a} = \pm\sqrt{\frac{b^2 - 4ac}{4a^2}}$$

$$x = \frac{-b}{2a} \pm \frac{\sqrt{b^2 - 4ac}}{2a}$$

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

Page 15 questions

Use the quadratic formula to solve the following quadratic equations, leaving your answers in surd form:

a $x^2 - 3x - 18 = 0$

$a = 1,\ b = -3,\ c = -18$

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

$$x = \frac{-(-3) \pm \sqrt{(-3)^2 - 4 \times 1 \times -18}}{2 \times 1}$$

$$x = \frac{3 \pm \sqrt{9 + 72}}{2}$$

$$x = \frac{3 \pm \sqrt{81}}{2}$$

$$x = \frac{3 \pm 9}{2}$$

$x = 6$ or $x = -3$

b $x^2 - 9x + 14$

$a = 1,\ b = -9,\ c = 14$

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

$$x = \frac{-(-9) \pm \sqrt{(-9)^2 - 4 \times 1 \times 14}}{2 \times 1}$$

$$x = \frac{9 \pm \sqrt{81 - 56}}{2}$$

$$x = \frac{9 \pm \sqrt{25}}{2}$$

$$x = \frac{9 \pm 5}{2}$$

$x = 7$ or $x = 2$

c $2x^2 + 26x + 80 = 0$

$a = 2,\ b = 26,\ c = 80$

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

$$x = \frac{-26 \pm \sqrt{26^2 - 4 \times 2 \times 80}}{2 \times 2}$$

$$x = \frac{-26 \pm \sqrt{676 - 640}}{4}$$

$$x = \frac{-26 \pm \sqrt{36}}{4}$$

$x = -8$ or $x = -5$

d $-4tx^2 + 32t - 60 = 0$

$a = -4,\ b = 32,\ c = -60$

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

$$x = \frac{-32 \pm \sqrt{32^2 - 4(-4)(-60)}}{2(-4)}$$

$$x = \frac{-32 \pm \sqrt{1024 - 960}}{-8}$$

$$x = \frac{-32 \pm \sqrt{64}}{-8}$$

$$x = \frac{-32 \pm 8}{-8}$$

$x = 5$ or $x = 3$

e $y^2 + 2y - 5 = 0$

$a = 1,\ b = 2,\ c = -5$

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

$$x = \frac{-2 \pm \sqrt{2^2 - 4(1)(-5)}}{2(1)}$$

$$x = \frac{-2 \pm \sqrt{4 + 20}}{2}$$

$$x = -1 \pm \sqrt{\frac{24}{4}}$$

$$x = -1 \pm \sqrt{6}$$

$x = -1 + \sqrt{6}$ or $x = -1 - \sqrt{6}$

f $-p^2 - 5p + 3 = 0$

$a = -1,\ b = -5,\ c = 5$

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

$$x = \frac{-(-5) \pm \sqrt{(-5)^2 - 4(-1)(5)}}{2(-1)}$$

$$x = \frac{5 \pm \sqrt{25 + 20}}{-2}$$

$$x = \frac{\pm\sqrt{45}}{-2}$$

$$x = -\frac{5}{2} \pm \frac{\sqrt{5 \times 9}}{-2}$$

$x = -\frac{5}{2} + \frac{3\sqrt{5}}{2}$ or $x = -\frac{5}{2} - \frac{3\sqrt{5}}{2}$

Page 16 questions

Use the quadratic formula to solve the following quadratic equations, leaving your answers in surd form:

a $-3x^2 - 10x + 8 = 0$

$a = -3,\ b = -10,\ c = 8$

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

$$x = \frac{-(-10) \pm \sqrt{(-10)^2 - 4(-3)(8)}}{2(-3)}$$

$$x = \frac{10 \pm \sqrt{100 + 96}}{-6}$$

$$x = \frac{10 \pm \sqrt{196}}{-6}$$

$x = -4$ or $x = \frac{2}{3}$

b $4x^2 - 9 = 0$

$a = 4,\ b = 0,\ c = -9$

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

$$x = \frac{0 \pm \sqrt{0 - 4(4)(-9)}}{2(4)}$$

$$x = \frac{\pm\sqrt{0144}}{8}$$

$x = \frac{3}{2}$ or $x = -\frac{3}{2}$

c $x(2x - 7) = -4$

$2x^2 - 7x + 4 = 0$

$a = 2,\ b = -7,\ c = 4$

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

$$x = \frac{-(-7) \pm \sqrt{(-7)^2 - 4(2)(4)}}{2(2)}$$

$$x = \frac{7 \pm \sqrt{49 - 32}}{4}$$

$x = \frac{7 + \sqrt{17}}{4}$ or $x = \frac{7 - \sqrt{17}}{4}$

d $x(4 - 5x) = -6$

$-5x^2 + 4x + 6 = 0$

$a = -5,\ b = 4,\ c = 6$

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

$$x = \frac{-4 \pm \sqrt{(4)^2 - 4(-5)(6)}}{2(-5)}$$

$$x = \frac{-4 \pm \sqrt{136}}{-10}$$

$$x = \frac{-4 \pm \sqrt{34 \times 4}}{-10}$$

$x = \frac{-4 + 2\sqrt{34}}{-10}$ or $x = \frac{-4 - 2\sqrt{34}}{-10}$

Page 17 questions

Solve $7x^2 + 42x - 112 = 0$ using each of the following method.
(You should get the same solutions each time)

a Factorise and solve for x

$$7x^2 + 42x - 112 = 0$$
$$7[x^2 + 6x - 16] = 0$$
$$7[(x-2)(x+8)] = 0$$
$$x - 2 = 0 \text{ or } x + 8 = 0 \text{ (by Null Factor Law)}$$
$$x = 2 \text{ or } x = -8$$

b Use completing the square to solve for x

$$7x^2 + 42x - 112 = 0$$
$$7[x^2 + 6x - 16] = 0$$
$$(x+3)^2 - 3^2 - 16 = 0$$
$$(x+3)^2 - 25 = 0$$
$$(x+3)^2 = 25$$
$$(x+3)^2 = (\pm 5)^2$$
$$x + 3 = \pm 5$$
$$x = 2 \text{ or } x = -8$$

c Use the quadratic formula to solve for x

$$7x^2 + 42x - 112 = 0$$
$$a = 7,\ b = 42,\ c = -112$$
$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$
$$x = \frac{-42 \pm \sqrt{42^2 - 4(7)(-112)}}{2(7)}$$
$$x = \frac{-42 \pm \sqrt{1764 + 3136}}{14}$$
$$x = \frac{-42 \pm \sqrt{4900}}{14}$$
$$x = \frac{-42 \pm \sqrt{70}}{14}$$
$$x = -3 \pm 5$$
$$x = -8 \text{ or } x = 2$$

Page 20 questions

Use the discriminant to determine if these equations have two equal, unequal or no real solutions
(Hint: Make sure the equation is in standard form $ax^2 + bx + c = 0$)

a $4x^2 + 21x - 18 = 0$

$a = 4,\ b = 21,\ c = -18$

$$\begin{aligned}\Delta &= b^2 - 4ac\\ &= 21^2 - 4(4)(-18)\\ &= 441 + 288\\ &= 729\end{aligned}$$

$\therefore \Delta > 0$ and is a perfect square

$\therefore$ the solutions will be unequaland rational

b $3x^2 + 7x + 2 = 0$

$a = 3,\ b = 7,\ c = 2$

$$\begin{aligned}\Delta &= b^2 - 4ac\\ &= 49 - 4(3)(2)\\ &= 25\end{aligned}$$

$\therefore \Delta > 0$ and is a perfect square

$\therefore$ the solutions will be unequal and rational

c $x^2 - 3x + 11 = 0$

$a = 1,\ b = -3,\ c = 11$

$$\begin{aligned}\Delta &= b^2 - 4ac\\ &= (-3)^2 - 4(1)(11)\\ &= 9 - 44\\ &= -35\end{aligned}$$

$\therefore \Delta < 0$

$\therefore$ there are no real solutions for x

d $2x^2 - 18 = 16x$

$2x^2 - 16x - 18 = 0$

$a = 2,\ b = -16,\ c = -18$

$$\begin{aligned}\Delta &= b^2 - 4ac\\ &= (-16)^2 - 4(2)(-18)\\ &= 256 + 144\\ &= 400 = 20^2\end{aligned}$$

$\therefore \Delta > 0$ and is a perfect square

$\therefore$ the solutions will be unequal and rational

e $x = \dfrac{3}{x} - 4$

$x^2 = 3 - 4x$

$x^2 + 4x - 3 = 0$

$a = 1,\ b = 4,\ c = -3$

$$\begin{aligned}\Delta &= b^2 - 4ac\\ &= 4^2 - 4(1)(-3)\\ &= 16 + 12\\ &= 28\end{aligned}$$

$\therefore \Delta > 0$ and is not a perfect square

$\therefore$ the solutions will be unequal and irrational

f $8x - 8 = 2x^2$

$-2x^2 + 8x - 8 = 0$

$a = -2,\ b = 8,\ c = -8$

$$\begin{aligned}\Delta &= b^2 - 4ac\\ &= 8^2 - 4(-2)(-8)\\ &= 64 - 64\\ &= 0\end{aligned}$$

$\therefore \Delta = 0$

$\therefore$ the two solutions will be equal.
These equal solutions will also be rational

Page 21 questions

Find the value of r if $rx^2 - 3r + 7 = 0$

a Has one solution

$rx^2 - 3x + 9 = 0$
$a = r,\ b = -3,\ c = 9$

if there is one solution then $\Delta = b^2 - 4ac = 0$
so $(-3)^2 - 4(r)(9) = 0$
$9 - 36r = 0$
$r = \frac{1}{4}$

$rx^2 - 3x + 9 = 0$ has one solution when $r = \frac{1}{4}$

b Has two unequal solutions

$rx^2 - 3x + 9 = 0$
$a = r,\ b = -3,\ c = 9$

for two unequal solutions $\Delta = b^2 - 4ac > 0$
$(-3)^2 - 4(r)(9) > 0$
$9 - 36r > 0$
$r < \frac{-9}{-36}$ (reverse the sign of the inequality when dividing by a negative number)
$r < \frac{1}{4}$

The equation $rx^2 - 3x + 9 = 0$ has unequal solutions when $r < \frac{1}{4}$

Find the values for k such that $4x^2 + 3kx - 3 = k + 6$ has no real solutions:

$x^2 + x + 7k = 6k - 3x$
$x^2 + 4x + k = 0$

there are no real solutions when $\Delta = b^2 - 4ac < 0$
$(4)^2 - 4(1)(k) < 0$
$-4k < -16$
$k > \frac{-16}{-4}$ (reverse the sign of the inequality when dividing by a negative number)
$k > 4$

$x^2 + x + 7k = 6k - 3x$ has no real solutions when $k > 4$

Page 22 questions

Use the discriminant to prove that $rx^2 + 2rx + r = 0$ has equal solutions for any r:

$$\begin{aligned}\Delta &= b^2 - 4ac \\ &= (2r)^2 - 4(r)(r) \\ &= 4r^2 - 4r^2 \\ &= 0\end{aligned}$$

Since $\Delta = b^2 - 4ac$ is always zero for $rx^2 + 2rx + r = 0$ it has equal solutions for any r

Prove that $2x^2 + 2kx + 2k^2 = 3kx - 3k^2$ never has real roots for any k:

$$2x^2 + 2kx + 2k^2 = 3kx - 3k^2$$

$$2x^2 - kx + 5k^2 = 0$$

$$a = 2,\ b = -k,\ c = 5k^2$$

$$\Delta = b^2 - 4ac$$

$$\Delta = (-k)^2 - 4(2)(5k^2)$$

$$\Delta = k^2 - 40k^2$$

$$\Delta = -39k^2$$

since k^2 must be positive, $-39k^2$ must be negative
So $\Delta = b^2 - 4ac < 0$ which means there are no real roots for any value of k

Page 22 questions

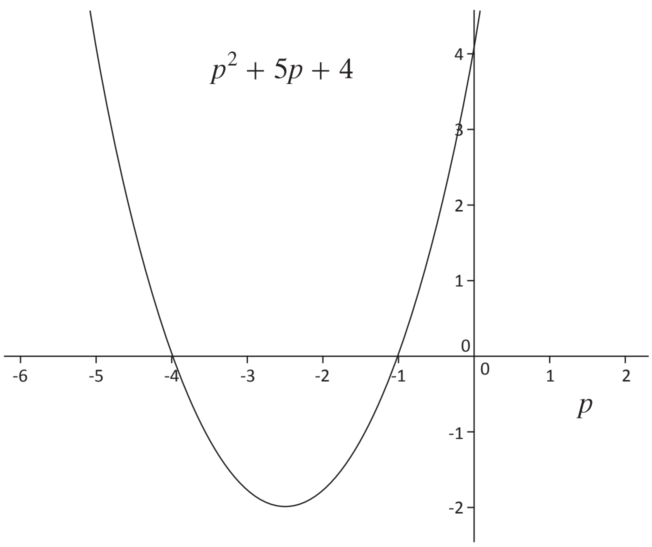
Find p if the following equation has two unequal real solutions:

$$\frac{2}{p} = \frac{2-2x^2}{5-4x}$$

$$p(2-2x^2) = 2(5-4x)$$
$$2p - 2px^2 = 10 - 8x$$
$$-2px^2 + 8x + 2p - 10 = 0$$
$$a = -2p,\ b = 8,\ c = 2p - 10$$

$$\Delta = b^2 - 4ac > 0$$
$$8^2 - 4(-2p)(2p-10) > 0$$
$$64 + 16p^2 + 80p > 0$$
$$p^2 + 5p + 4 > 0$$
$$(p+1)(p+4) > 0$$

we can plot this to interpret the points $p = -1$ and $p = -4$



Plotting the discriminant function, $p^2 + 5p + 4$, we can see it is positive, when $p < -4$ or $p > -1$

So $\frac{2}{p} = \frac{2-2x^2}{5-4x}$ has two unequal and real solutions when $p < -4$ or $p > -1$

Page 24 questions

Substitute $k = x^3$ to solve for x in $2x^6 + 4x^3 - 16 = 0$

Let $k = x^3$

$\therefore 2k^2 + 4k - 16 = 0$

$\therefore (2k-4)(k+4) = 0$

$\therefore 2k - 4 = 0$ or $k = -4$

$\therefore k = 2$ or $k = -4$

$\therefore x^3 = 2$ or $x^3 = -4$

$\therefore x = \sqrt[3]{2}$ or $x = \sqrt[3]{-4}$

Make a substitution to solve for x in

a $40 + 6(7x+3) - (7x+3)^2 = 0$

let $y = 7x + 3$

$40 + 6y - y^2 = 0$

$y^2 - 6y - 40 = 0$

$(y+4)(y-10) = 0$

$y = -4$ or $y = 10$

$7x + 3 = -4$ or $7x + 3 = 10$

$x = -1$ or $x = 1$

b $(x-2)^4 - 5(x-2)^2 - 36 = 0$

$let\ a = (x-2)^2$

$a^2 - 5a - 36 = 0$

$(a-9)(a+4) = 0$

$a = 9\ or\ a = -4$

$(x-2)^2 = 9\ or\ (x-2)^2 = -4$ This is impossible

$(x-2)^2 = (\pm 3)^2\ or\ (x-2)^2 = (\pm\sqrt{-4})^2$

c $9^x - 10(3^x) + 9 = 0$ (Hint $9 = 3^2$)

$(3^x)^2 - 10(3x) + 9 = 0$ since $9^x = (3^2)^x = 3^{2x} = (3^x)^2$

let $b = 3^x$

$b^2 - 10b + 9 = 0$

$(b-9)(b-1) = 0$

$b = 9$ or $b = 1$

$3^x = 9$ or $3^x = 0$

$x = 2$ or $x = 0$

Page 26 questions

The product of two <u>consecutive</u> integers is 272

a Find the two numbers if they are positive.

Call the first number n

$$\begin{aligned} n(n+1) &= 272 \\ n^2+n-272 &= 0 \\ (n-16)(n+17) &= 0 \end{aligned}$$

so for positive numbers $n = 16$ and $n + 1 = 17$

(the numbers are 16 and 17)

b Find the two numbers if they are negative

$$\begin{aligned} n(n+1) &= 272 \\ n^2+n-272 &= 0 \\ (n-16)(n+17) &= 0 \end{aligned}$$

so for negative numbers $n = -17$ and $n + 1 = -16$

(the numbers are -17 and -16)

A rectangle's length is 3 less than four times the breadth. Find the dimensions of the rectangle if the area is 126 cm²:

Breadth: x
Length: $4x - 3$

$$\begin{aligned} \text{Area} = x \times (4x-3) &= 126 \\ 4x^2 - 3x &= 126 \\ 4x^2 - 3x - 126 &= 0 \\ x &= \frac{-b \pm \sqrt{b^2-4ac}}{2a} \\ &= \frac{-(-3) \pm \sqrt{(-3)^2 - 4(4)(-126)}}{2(4)} \\ &= \frac{3 \pm \sqrt{2025}}{8} \\ &= \frac{3 \pm 45}{2} \end{aligned}$$

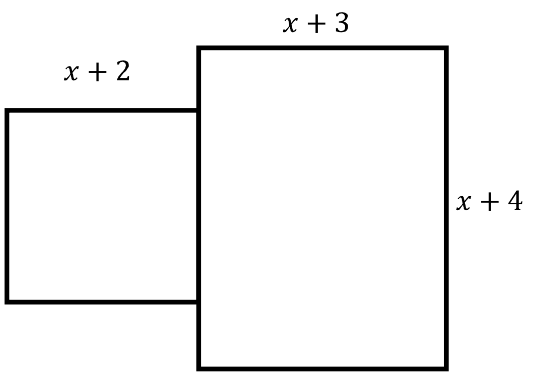
$x = 6$ or $x = -\frac{21}{4}$

Negative length doesn't make sense so breadth is 6 cm

Length is $4 \times 6 - 3 = 21$ cm

Page 27 questions

The stage below is made up of a square and a rectangle. Find x If the total area of the stage is $191\ \text{m}^2$



Total area = area of square + area of rectangle

$$(x+2)(x+2)+(x+3)(x+4)=191$$

$$x^2+4x+4+x^2+7x+12=191$$

$$2x^2+11x-175=0$$

$$(x-7)(2x+25)=0$$

$$x=7 \text{ or } x=-\frac{25}{2}$$

Negative length not possible so $x = 7\ \text{m}$

Page 30 questions

Solve these simultaneous equations for x and y. How many times does the straight line intersect the parabola?

a $y = x^2 + 13x + 1 \quad (1)$
$y = 8x + 1 \quad (2)$

$$x^2 + 13x + 1 = 8x + 1$$
$$x^2 + 5x = 0$$
$$x(x+5) = 0$$
$$x = 0 \text{ or } x = -5$$

Substitute $x = 0$ into (2)

$$y = 8(0) + 1$$
$$= 1 \text{ so } (0, 1) \text{ is one solution}$$

Substitute $x = -5$ into (2)

$$y = 8(-5) + 1$$
$$= -39 \text{ so } (-5, -39) \text{ is a solution}$$

The straight line intersects the parabola twice: at $(0, 1)$ and $(-5, -39)$

Page 31 questions

Solve these simultaneous equations for x and y. How many times does the straight line intersect the parabola?

b $y - 5 = 4x^2 - 10x \quad (1)$

$y + 3 = 2x \quad (2)$

$y = 4x^2 - 10x + 5 \quad (1)$

$y = 2x - 3 \quad (2)$

$4x^2 - 10x + 5 = 2x - 3$

$4x^2 - 12x + 8 = 0$

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

$$= \frac{-(-12) \pm \sqrt{(-12)^2 - 4(4)(8)}}{2(4)}$$

$$= \frac{12 \pm \sqrt{16}}{8}$$

$x = 2$ or $x = 1$

Substitute $x = 2$ into (2)

$y = 2(2) - 3$

$y = 1$ so $(2, 1)$ is a solution

Substitute $x = 1$ into (2)

$y = 2(1) - 3$

$y = -1$ so $(1, -1)$ is a solution

The straight line intersects the parabola twice: at $(2,\ 1)$ and $(1,\ -1)$

Page 32 questions

Solve these simultaneous equations for x and y. Find the coordinates of the point(s) where the graphs would intersect:

a
$$y - 3x = x^2 - 37 \qquad (1)$$
$$-23 = 7x - y \qquad (2)$$

$$y = x^2 + 3x - 37 \qquad (1)$$
$$y = 7x + 23 \qquad (2)$$

$$x^2 + 3x - 37 = 7x + 23$$
$$x^2 - 4x - 60 = 0$$
$$(x - 10)(x + 6) = 0$$
$$x = 10 \text{ or } x = -6$$

Substitute $x = 10$ into (2)

$$y = 7 \times 10 + 23$$
$= 93$ so $(10, 93)$ is one part of intersection

Substitute $x = -6$ into (2)

$$y = 7 \times (-6) + 23$$
$= -19$ so $(-6, -19)$ is one part of intersection

The graphs intersect at $(10,93)$ and $(-6, -19)$

Page 33 questions

Solve these simultaneous equations for x and y. Find the coordinates of the point(s) where the graphs would intersect:

a $y = 3x^2 - 9x + 30 \qquad (1)$

$y = -2x + 15 \qquad (2)$

$3x^2 - 9x + 30 = -2x + 15$

$3x^2 - 7x + 15 = 0$

$$\begin{aligned} x &= \frac{-b \pm \sqrt{b^2 - 4ac}}{2a} \\ &= \frac{-(-7) \pm \sqrt{(-7)^2 - 4(3)(15)}}{2(3)} \\ &= \frac{7 \pm \sqrt{-131}}{6} \end{aligned}$$

The discriminant is < 0 so this can't be solved (no real solutions)